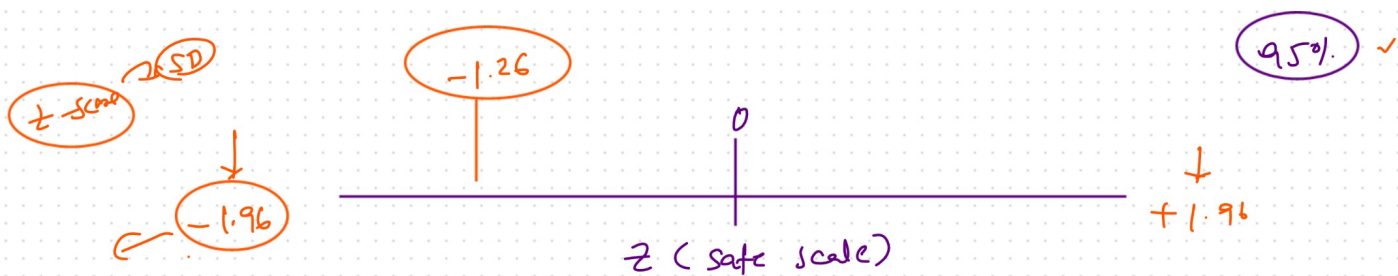
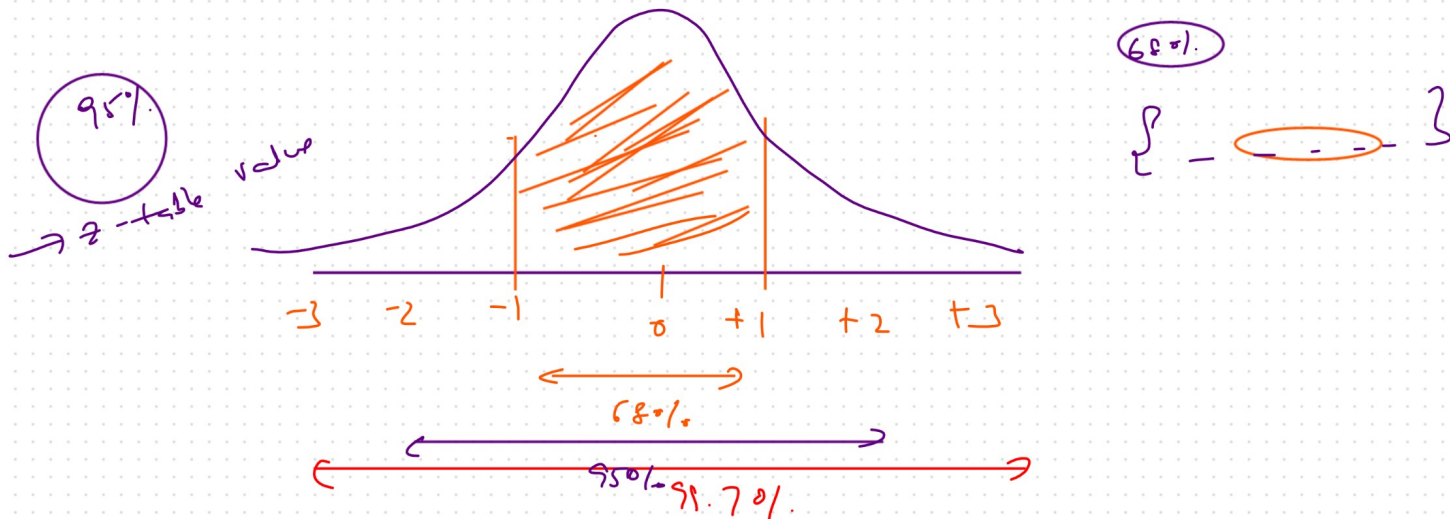
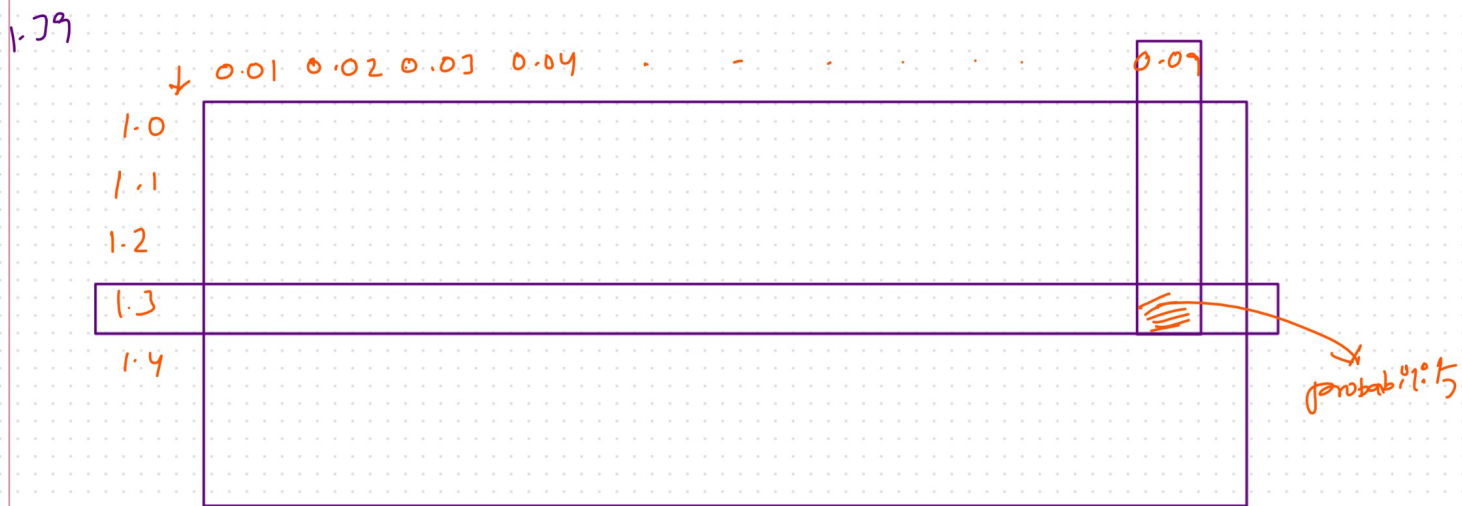
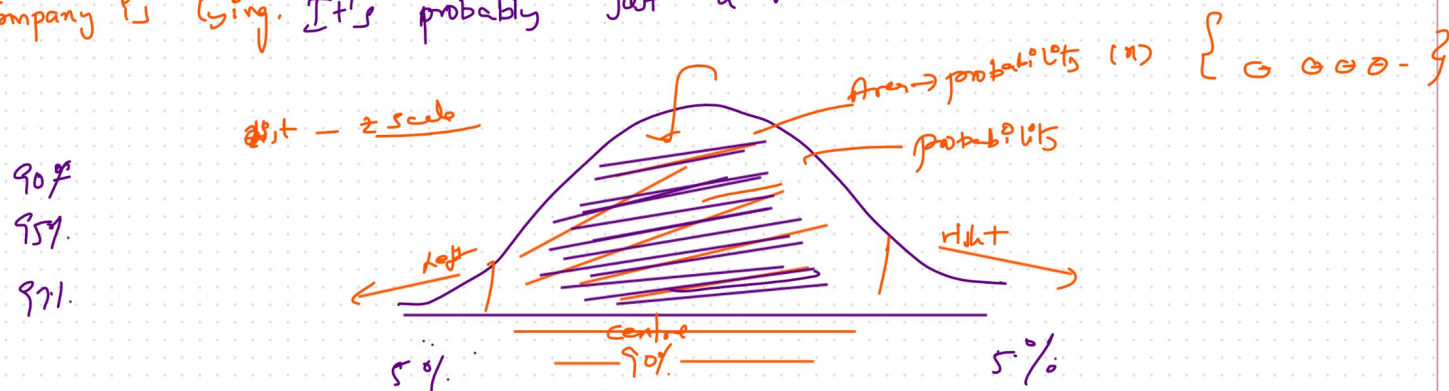


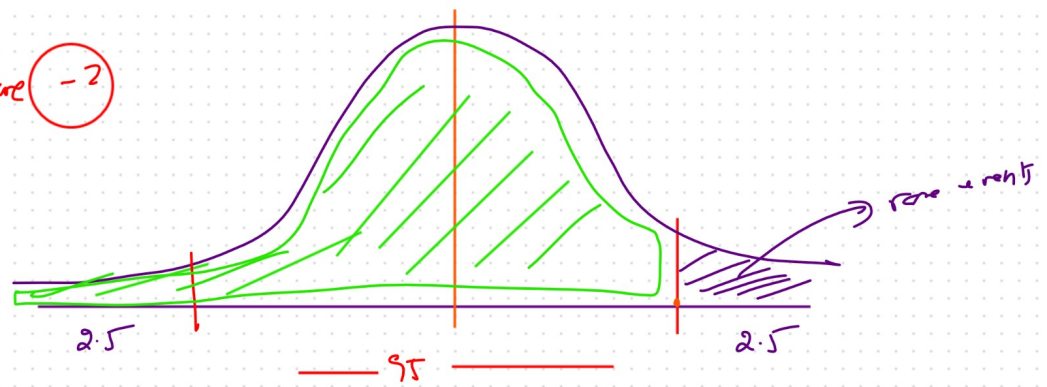
Statistics



1.98 is lower than 2SD, but is not weird enough to prove that company is lying. It's probably just a random coincidence



95% — z score (-2)



$$\text{left tail} + \text{center} = 2.5 + 95 = 97.5\% = 1.96$$

Significance level (α)

A — flips coin 2 times $\rightarrow$ 2 heads

— flips coin 20 times $\rightarrow$ 20 heads

$\nearrow$
This is too unlikely to be luck / happen

$\downarrow$
too rare

maths rule $\rightarrow$ significance level

$\rightarrow$ The significance level (α) is the probability of rejecting the claim when it is actually true

— Tolerance for Error

$\alpha = 0.05 \rightarrow 5\%$ $\rightarrow$ we are okay with being wrong

5% of the time

$\alpha = 0.01 (1\%)$ $\rightarrow$ we only want to be wrong 1% of the time

$\alpha = \underline{0.05} (5\%)$

95% sure



significance level

Two tailed

confidence level

$$\alpha = 0.05$$

$$z \rightarrow \pm 1.96$$

$$\rightarrow 95\%$$

$$\alpha = 0.01$$

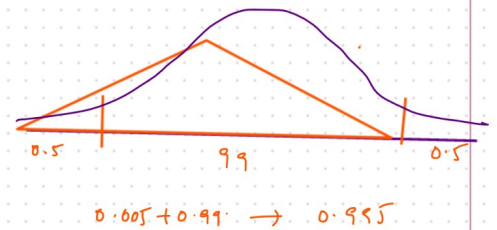
$$z \rightarrow \pm 2.58$$

$$\rightarrow 99\%$$

$$\alpha = 0.10$$

$$z \rightarrow \pm 1.645$$

$$\rightarrow 90\%$$



Hypothesis Testing

↓
Court

↓

innocent until proven guilty

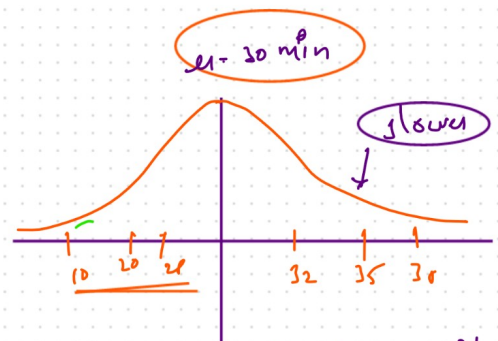
→ Null hypothesis (H_0) = "innocent state"
= "everything is normal"

→ Alternate hypothesis (H_1) = "Guilty state"
"Accusation"

Tails

Two-tailed test:

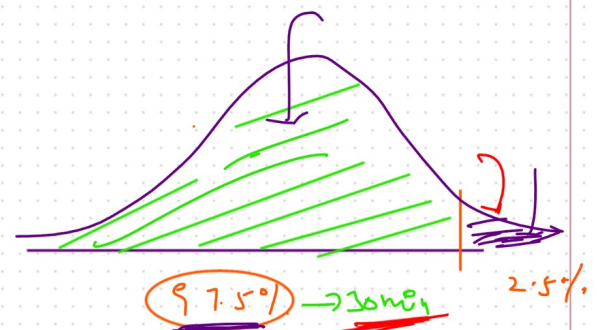
↓
30 minutes



∴ you don't care if it's faster or slower, you just want to know if it's not 30 minutes

5% significance → left → 2.5
right → 2.5

Critical value → ± 1.96



One-tailed test :

∴ The customer group specially claims you are slower

$$\downarrow$$
$$\textcircled{30} > 30 \rightarrow \text{Slower}$$

↓ we only care about right side of the curve

5% significance

$$\alpha = 0.05$$

0.95% confidence

Critical value = +1.64

$$\alpha = 1 - 0.95$$
$$= 0.05$$

$$1 - \alpha$$
$$= 1 - 0.05$$
$$= 95.00\%$$

P.S

The claim (H_0) : average delivery time ($\mu = \underline{30}$ minutes)

The Accusation (H_1) : A customer group say you are slower

They tested 36 deliveries ($n=36$)

Evidence : $\bar{x} = 31$ minutes

✓ Population standard deviation (σ) = 4 minutes

Significance level (α) = 0.05 (95% confidence)

Step 1 : State the hypothesis

$$H_0 : \mu = 30$$

$$H_1 : \mu > 30 \text{ (31)} \quad (\text{one tailed accusation})$$

Step 2 :

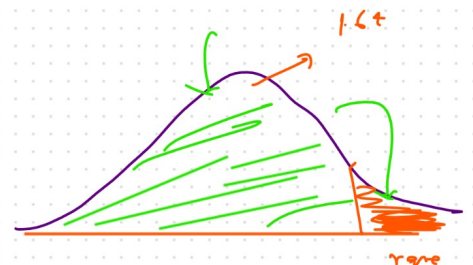
set the decision rule

$$\alpha = 0.05$$

z-score

group average

z-stats



$$Z\text{-statistic} > 1.64 \longrightarrow \text{"Guilty"}$$

↓
reject H_0

H_1

pass
failed to reject H_0

rejected to
failed

Step 3: Calculate SE

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{36}} = 0.667$$

Step 4: Calculate Z-statistic

$$Z_{\text{stat}} = \frac{\bar{x} - \mu}{SE} = \frac{31 - 30}{0.667} = \frac{1}{0.667} \approx 1.50$$

$$Z_{\text{stat}} = 1.50$$

$$\text{is } 1.50 > 1.64 \longrightarrow \text{No} \quad \underline{1.72}$$

Conclusion: Failed to reject Null Hypothesis

Even though the average delivery time was 31 minutes, a sample of 36 is small enough that this could have been a bad week [By random chance]
we don't have enough proof to call the company

P-value

A → flipped coin 2 times → 2 head

flipped coin 10 times → 10 head

0.1% chance → how sure we are

↑
P-value

P-value — lower p-value means rare
 — higher p-value means common

[0, 1]

Rule : if the p-value is low, then the Null hypothesis is rejected

(p-value $< \alpha$, we reject the H_0)

Example

$$H_0 = \mu = 30$$

$$H_1 = \mu > 30 \text{ (one tailed)}$$

$$\alpha = 0.05$$

→ taken from previous example

$$z\text{-stats} = 1.5$$

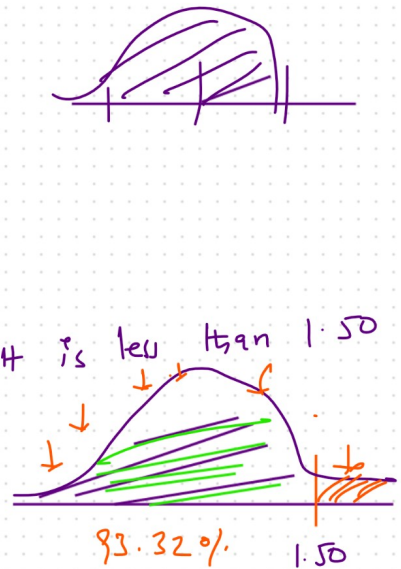
$$P(z > 1.50)$$

$$z = 1.50 \rightarrow \text{Area?}$$

$$\rightarrow 0.93319$$

$$= 0.9332$$

→ 93.32% chance the result is less than 1.50



$$P\text{-value} = 1 - \text{Area to the left}$$
$$= 1 - 0.9332$$

$$= 0.0668$$

$$\alpha = 0.05$$

$$0.0668 < 0.05$$

$$\underline{6.68\%} < 5\%$$

93.3% 6.6%

There's a 6.68% chance that 31-minute delivery average was a random fluke

Cup - -

⇒ Not statistically significant